Create and mount an Amazon EFS file system

using Amazon EC2 Launch Instance Wizard

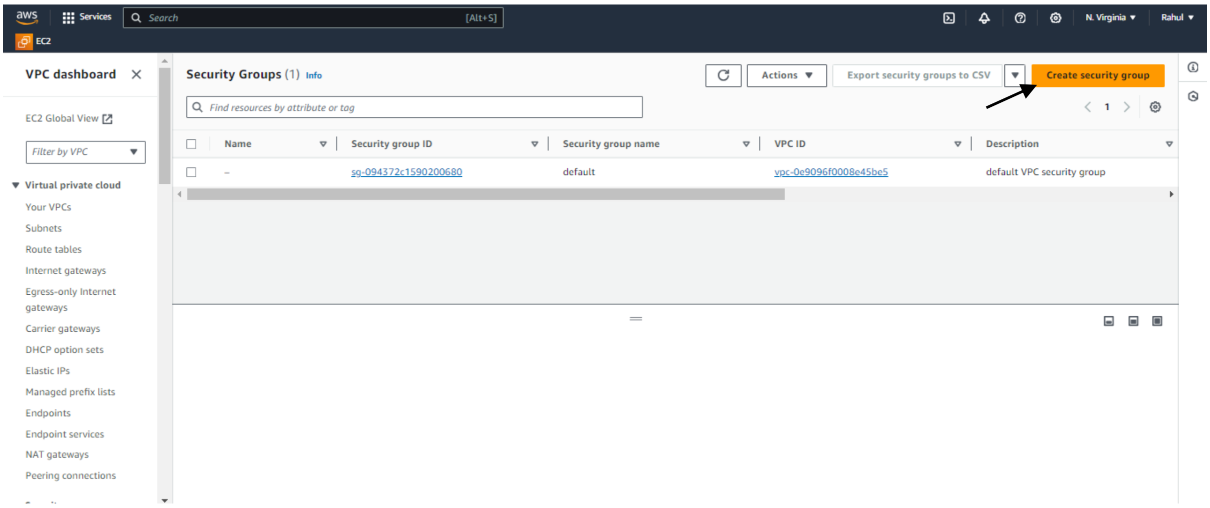
Step 1: Create the security groups

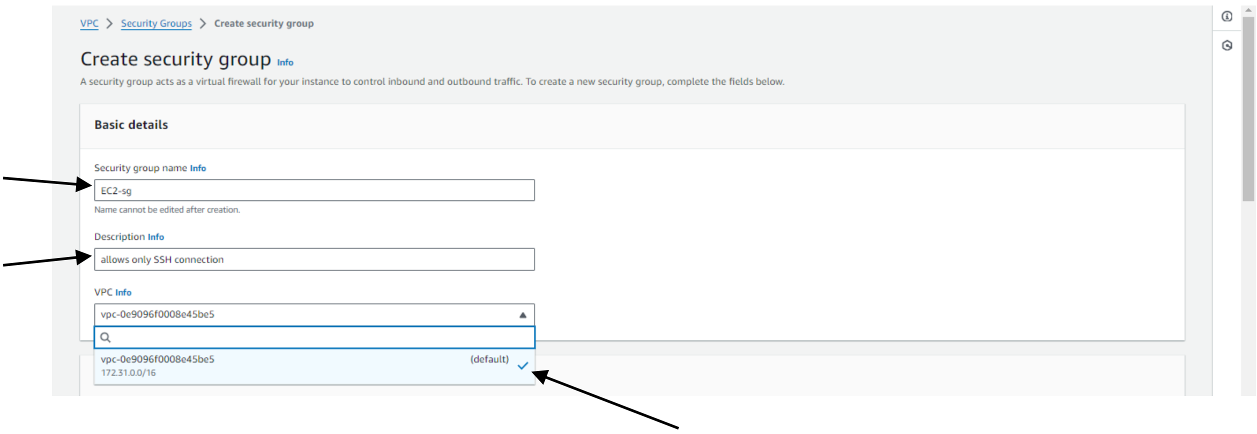
For this step, you create two security groups - one that is attached to the Amazon EC2 instance and one that is attached to the Amazon EFS file system.

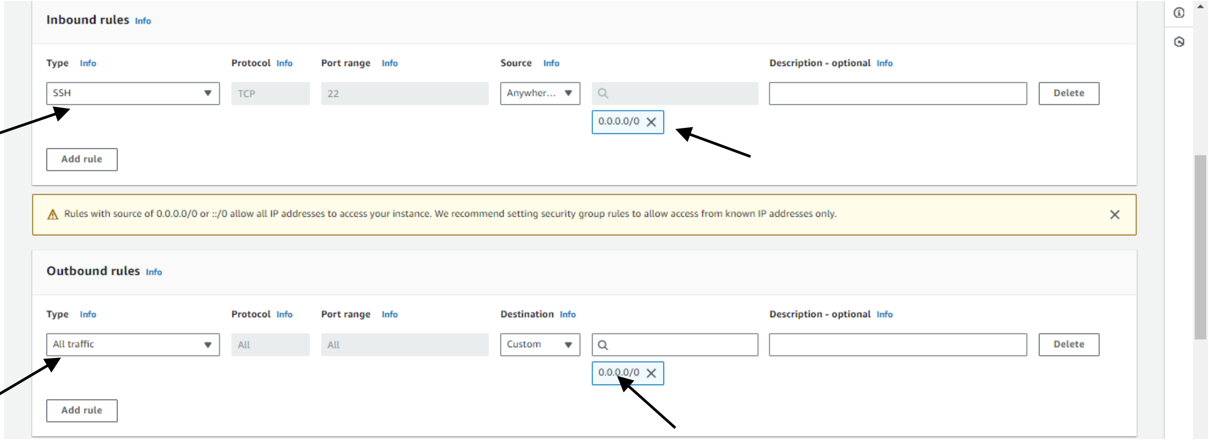
Open the [Amazon VPC dashboard] and In the left navigation pane, choose Security Groups. There will be a default security group listed. Create the following two additional security groups in the default VPC. For detailed steps, see [Creating a Security Group].

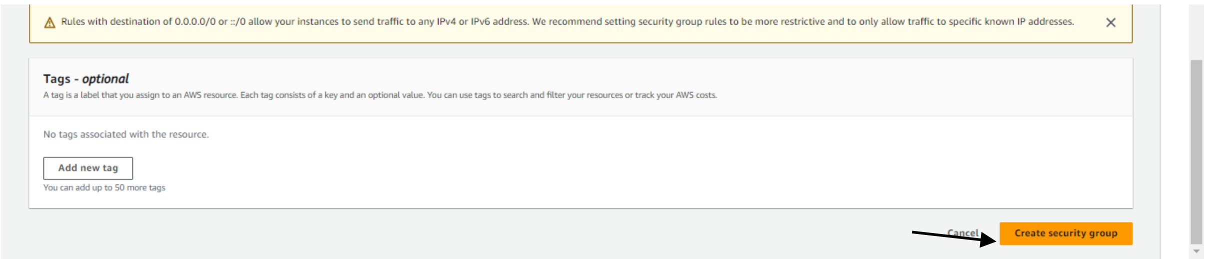
- EC2-sg: This security group will be attached to the EC2 instance and it allows only SSH connection inbound to the EC2 instance and any outbound connectivity.
- EFS-sg: This security group will be attached to the EFS file system and allows only TCP connection on port 2409 from the EC2 instance and any outbound connectivity.

Security Group1: -

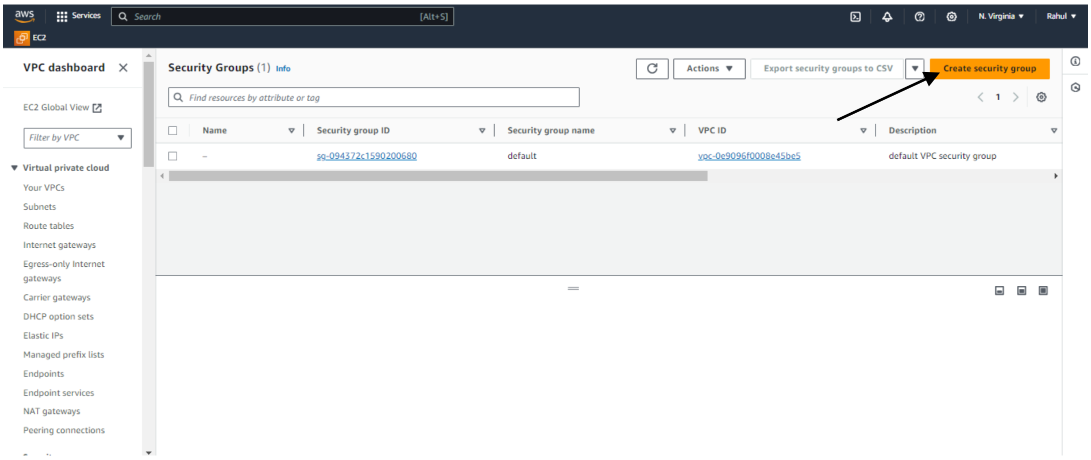


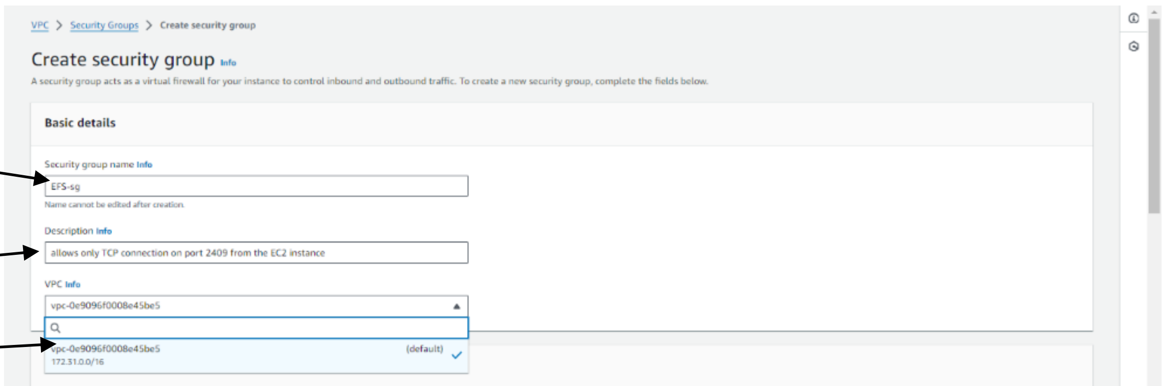




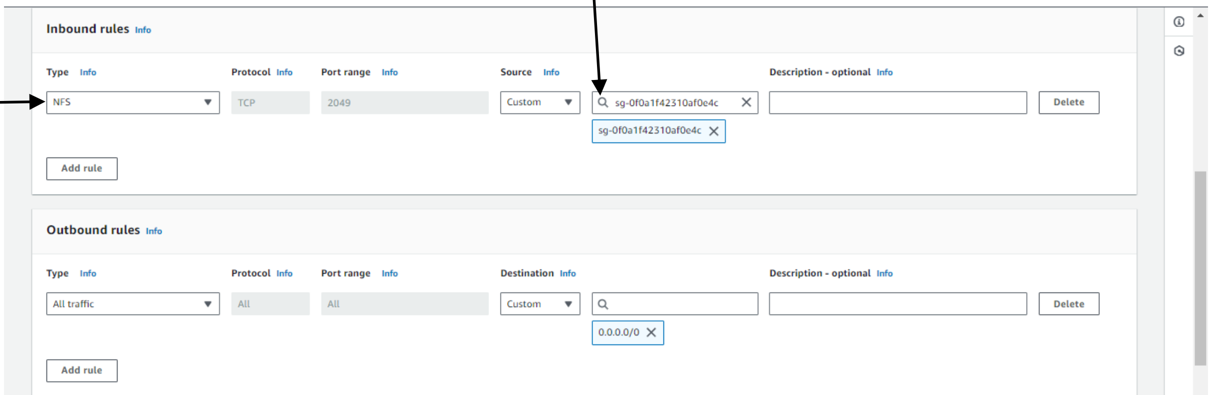


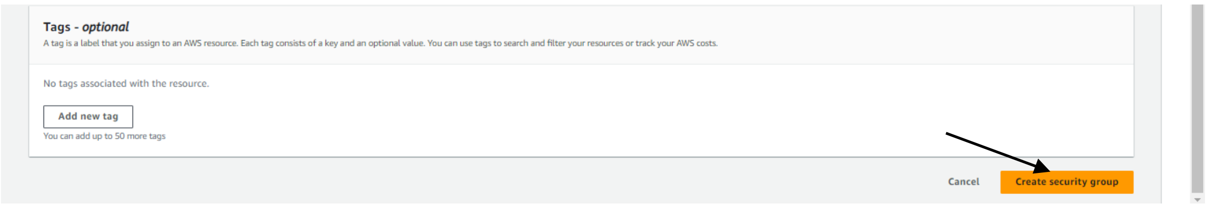
Security Group2: -

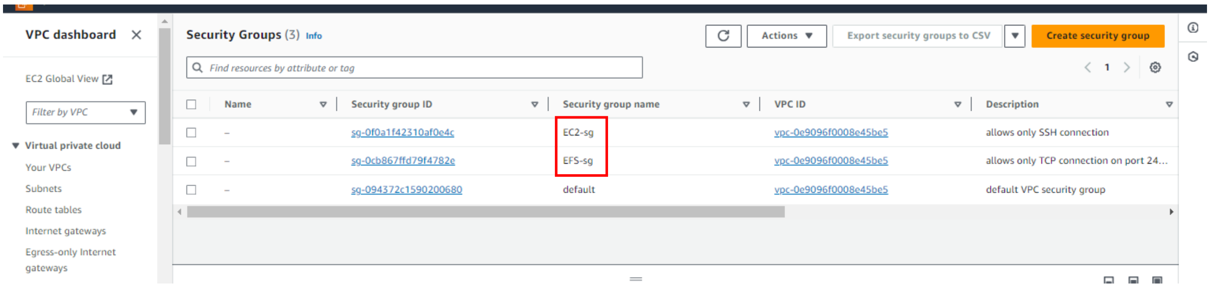




Select EC2-sg security group

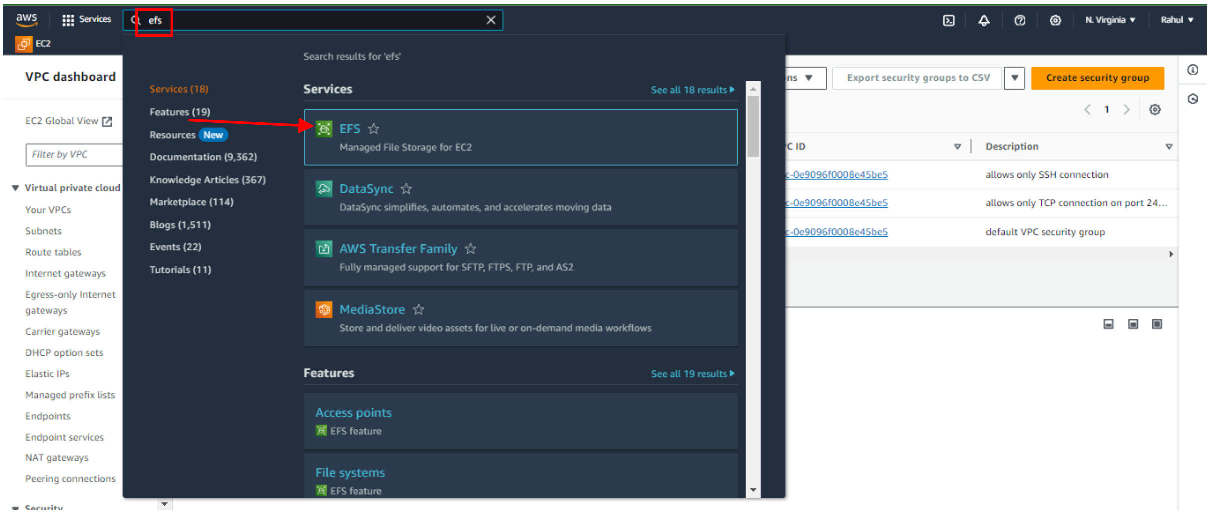


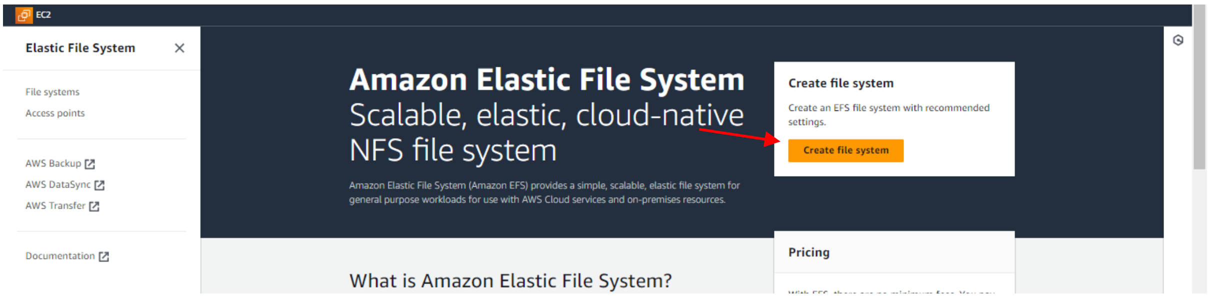


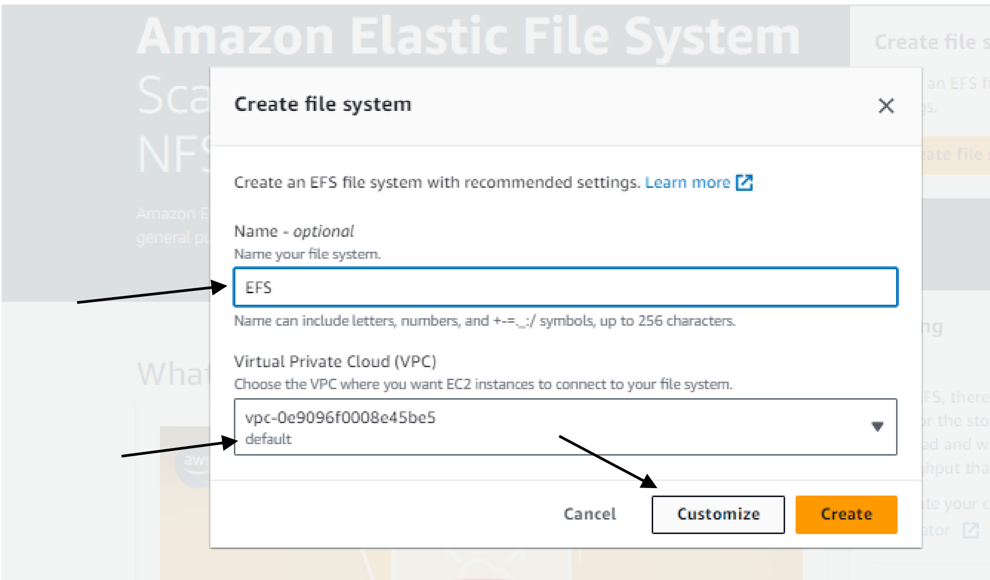


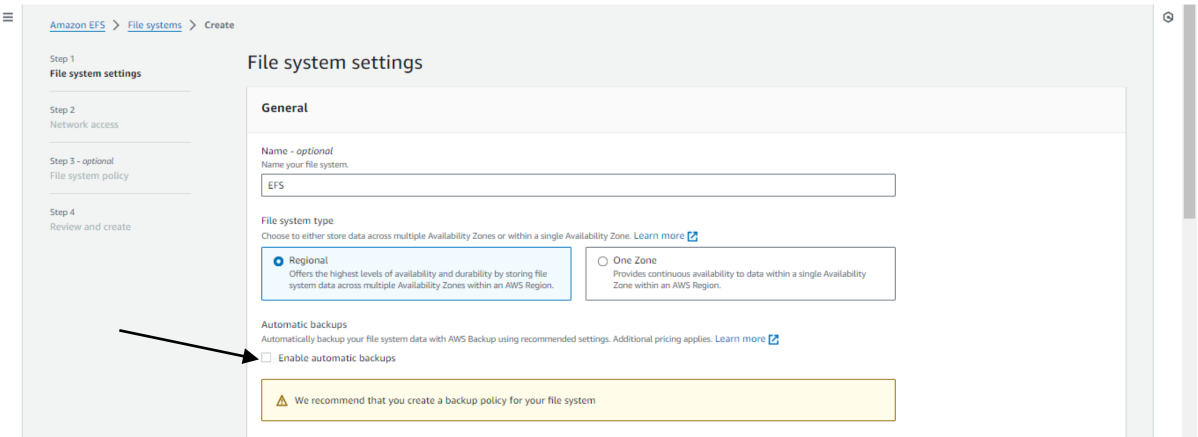
Step 2: Create the EFS file system

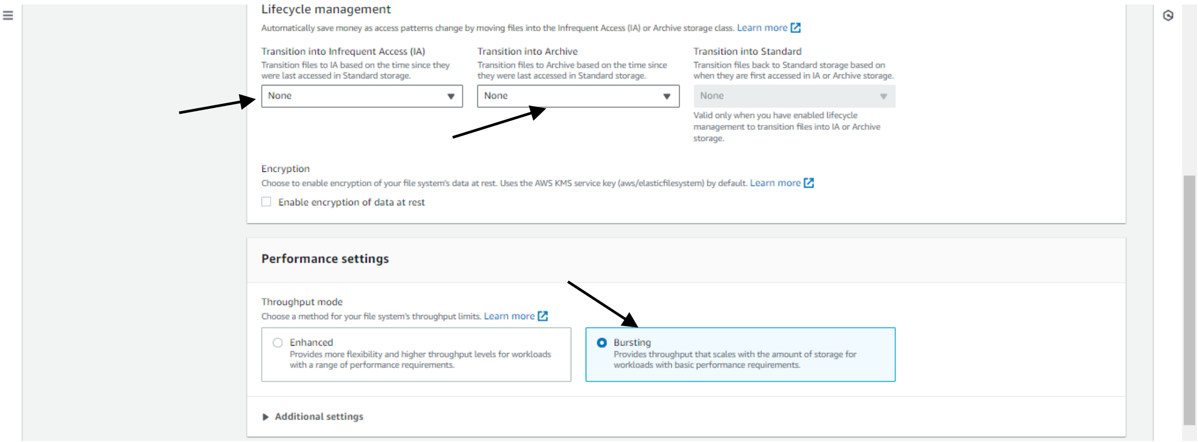
Navigate to [Amazon EFS in the AWS Management Console]() and choose Create file system.

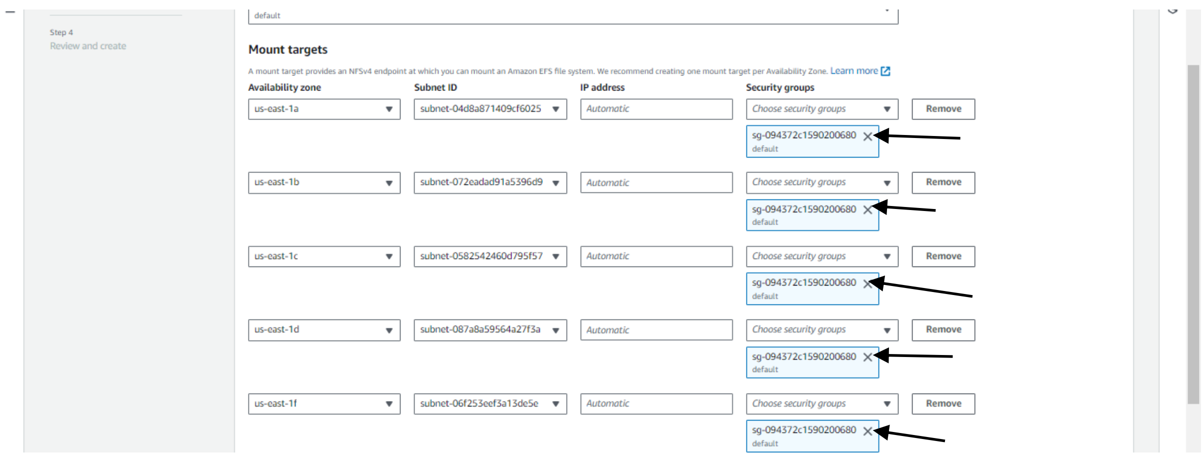




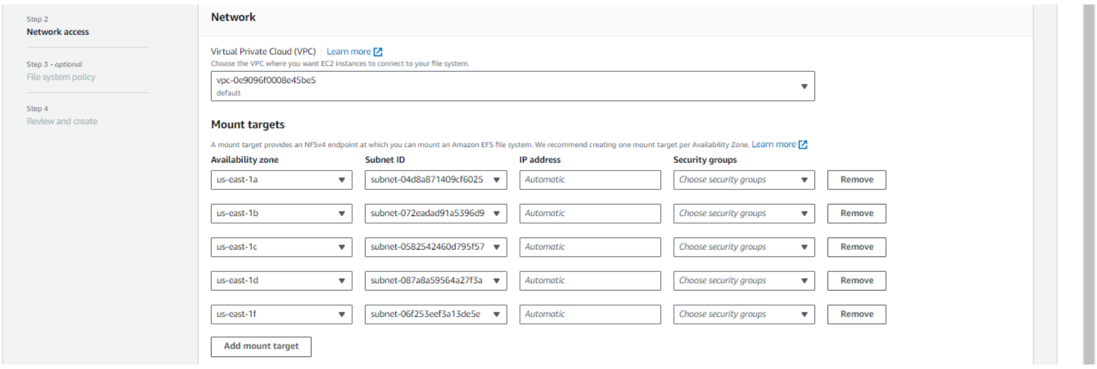


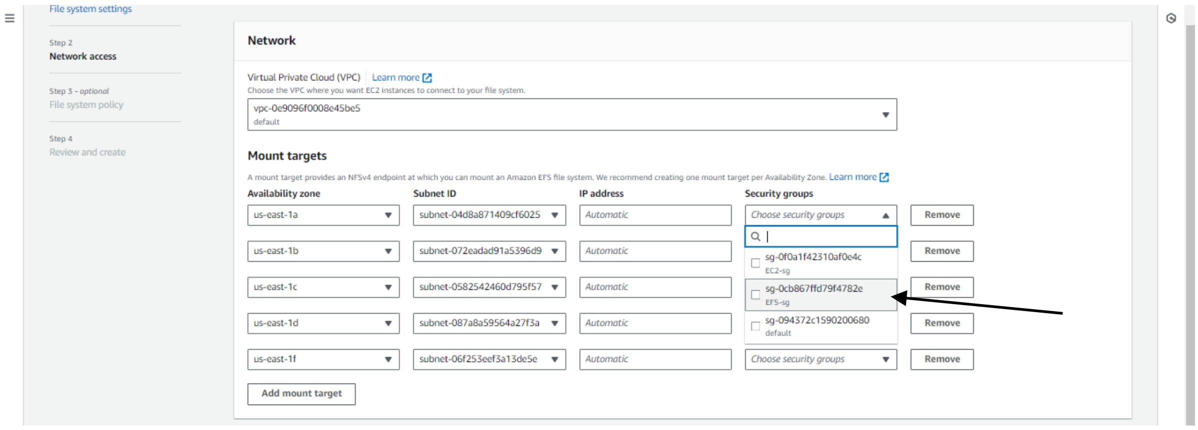






In the Security groups column, delete the existing security groups and add the EFS-sg security group.


Step 2
Network access
Step 3 - optional
File system policy
Step 4
Review and create
Network
Virtual Private Cloud (VPC) Learn more
Choose the VPC where you want EC2 instances to connect to your file system.
vpc-0e9096f0008e45be5
default
Mount targets
A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. Learn more
Availability zone
Subnet ID
IP address
Security groups
us-east-1a
subnet-04d8a871409cf6025
Automatic
Choose security groups
Remove
us-east-1b
subnet-072eadad91a5396d9
us-east-1c
subnet-0582542460d795f57
us-east-1d
subnet-087a8a59564a27f3a
us-east-1f
subnet-06f253eef3a13de5e
Add mount target


File system settings
Step 2
Network access
Step 3 - optional
File system policy
Step 4
Review and create
Network
Virtual Private Cloud (VPC) Learn more
Choose the VPC where you want EC2 instances to connect to your file system.
vpc-0e9096f0008e45be5
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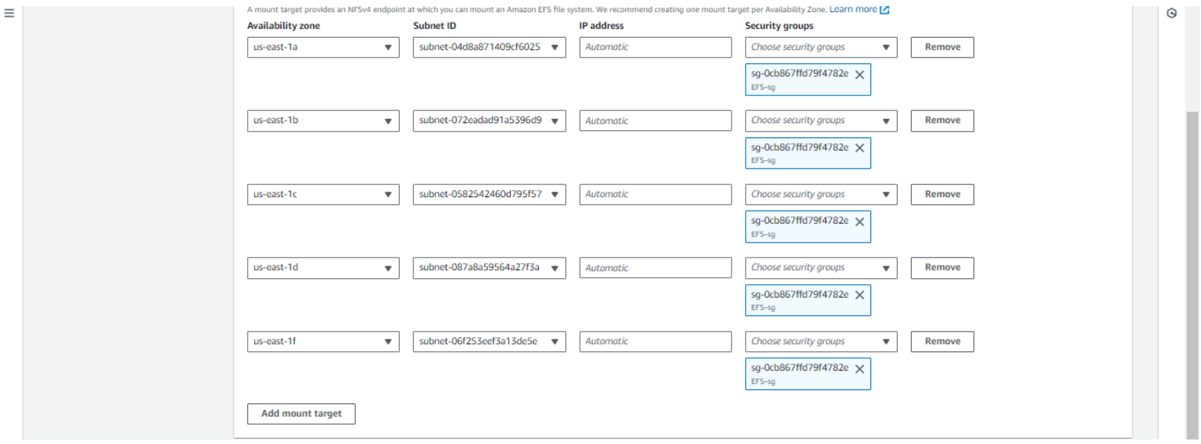
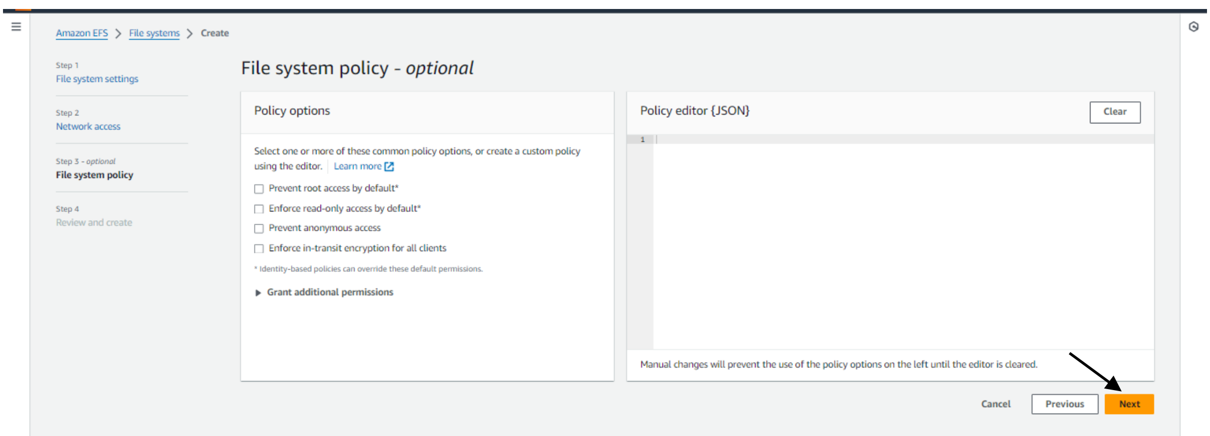
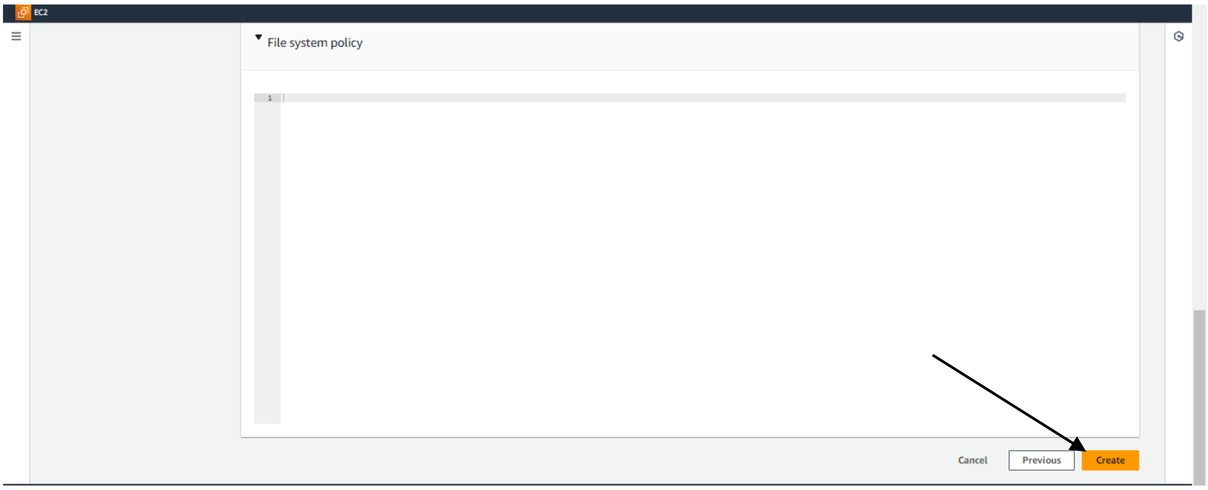
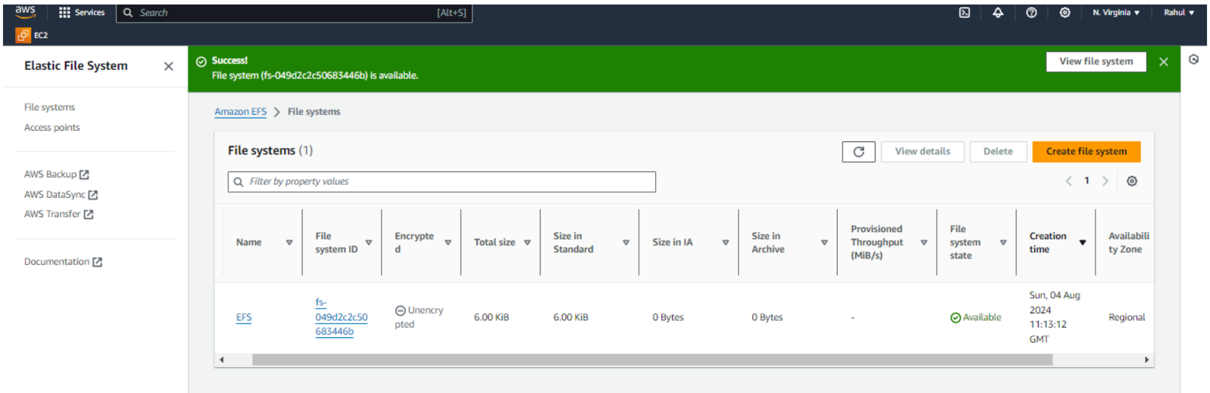
Availability zone
Subnet ID
IP address
Security groups
us-east-1a
subnet-04d8a871409cf6025
Automatic
Choose security groups
Remove
sg-0f0a1f42310af0e4c
EC2-sg
sg-0cb867ffd79f4782e
EFS-sg
sg-094372c1590200680
default
us-east-1b
subnet-072eadad91a5396d9
us-east-1c
subnet-0582542460d795f57
us-east-1d
subnet-087a8a59564a27f3a
us-east-1f
subnet-06f253eef3a13de5e
Add mount target


A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. Learn more
Availability zone
Subnet ID
IP address
Security groups
us-east-1a
subnet-04d8a871409cf6025
Automatic
Choose security groups
Remove
sg-0cb867ffd79f4782e
EFS-sg
us-east-1b
subnet-072eadad91a5396d9
us-east-1c
subnet-0582542460d795f57
us-east-1d
subnet-087a8a59564a27f3a
us-east-1f
subnet-06f253eef3a13de5e
Add mount target


Amazon EFS > File systems > Create
Step 1
File system settings
Step 2
Network access
Step 3 - optional
File system policy
Step 4
Review and create
File system policy - optional
Policy options
Select one or more of these common policy options, or create a custom policy using the editor. Learn more
Prevent root access by default*
Enforce read-only access by default*
Prevent anonymous access
Enforce in-transit encryption for all clients
* Identity-based policies can override these default permissions.
Grant additional permissions
Policy editor {JSON}
Clear
1
Manual changes will prevent the use of the policy options on the left until the editor is cleared.
Cancel
Previous
Next

Under Review and create

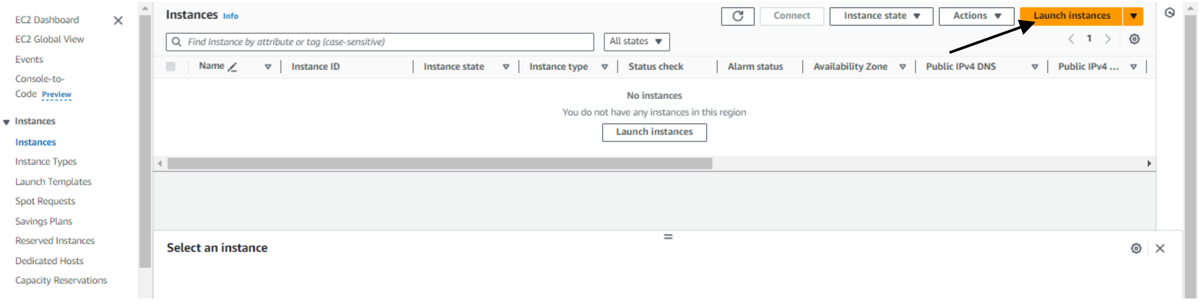


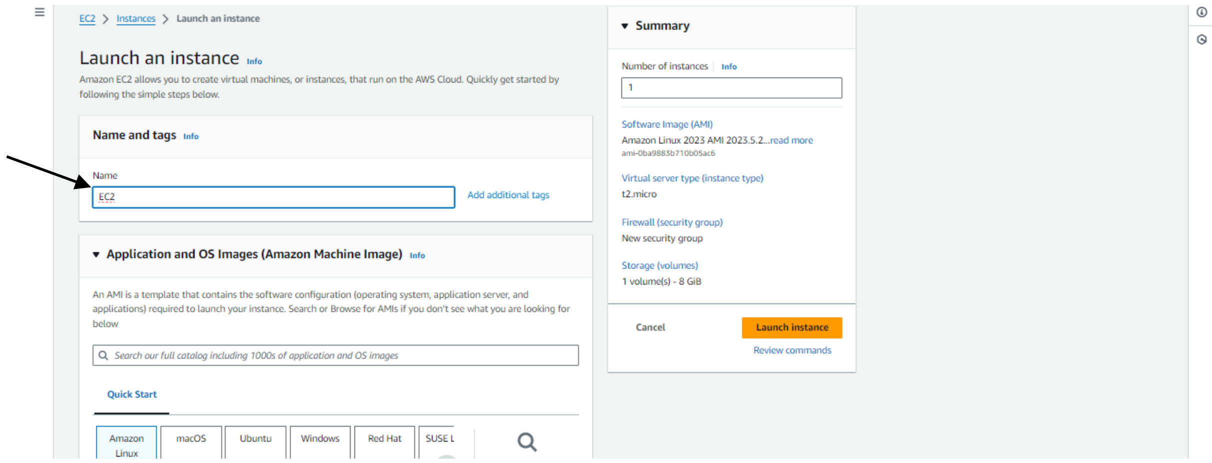


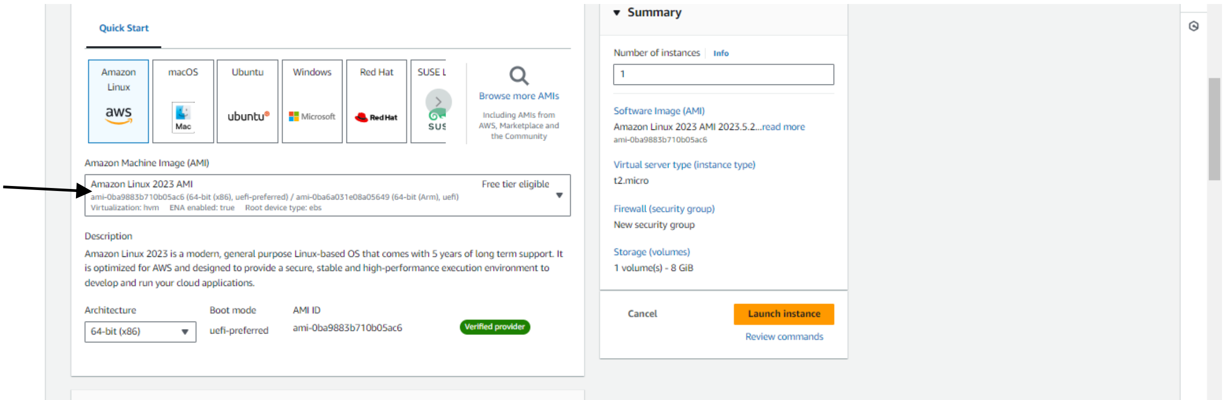
Successfully Created!!

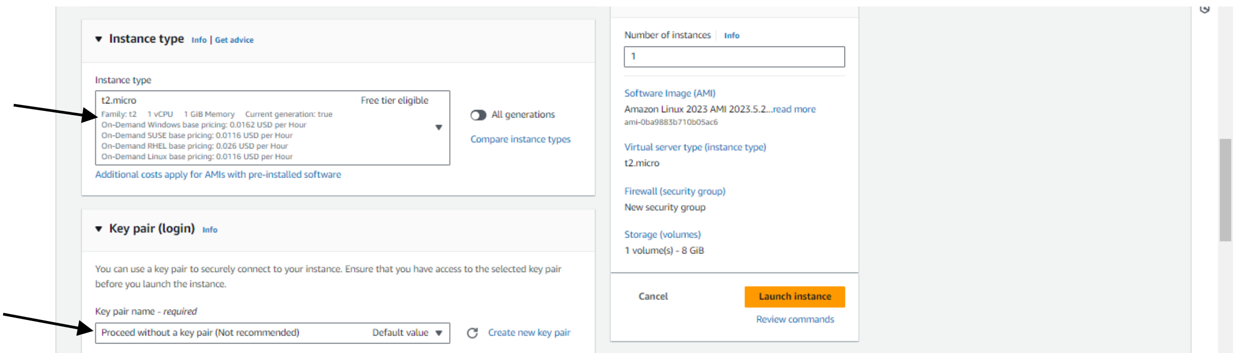
Step 3: Launch the EC2 instance and mount the file system

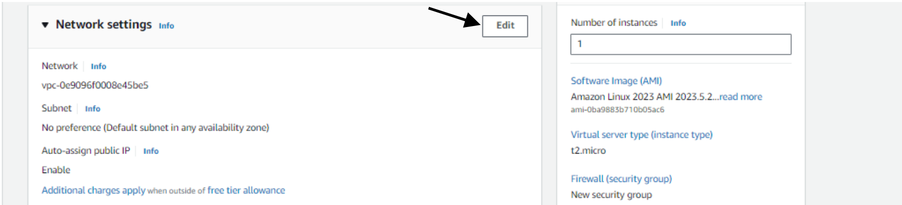
Navigate to the Amazon EC2 console and choose Launch Instance.



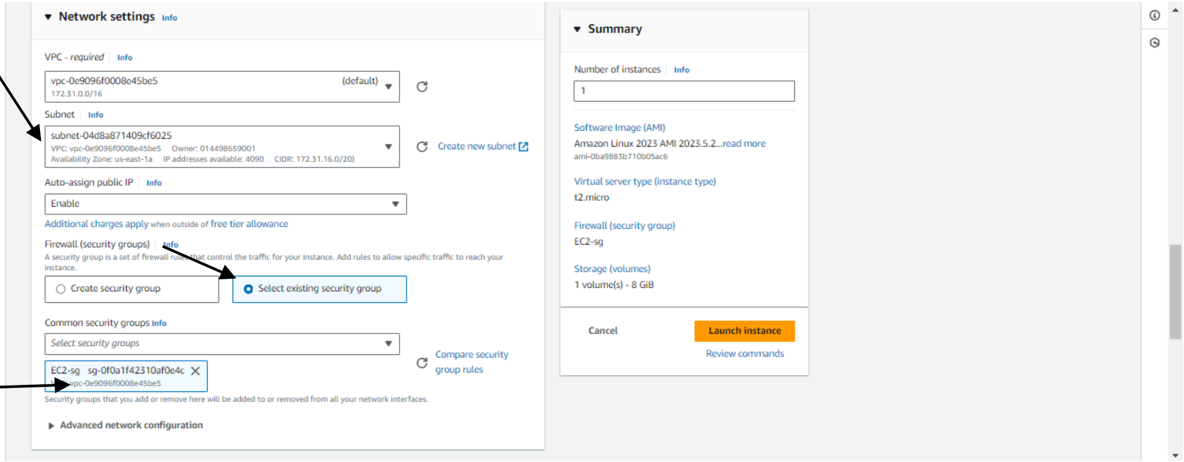


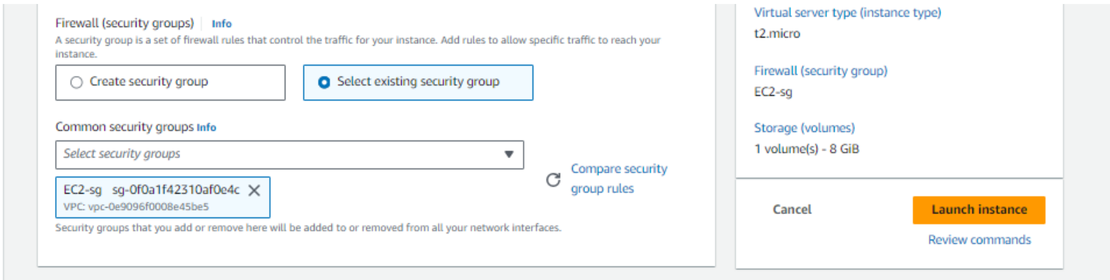


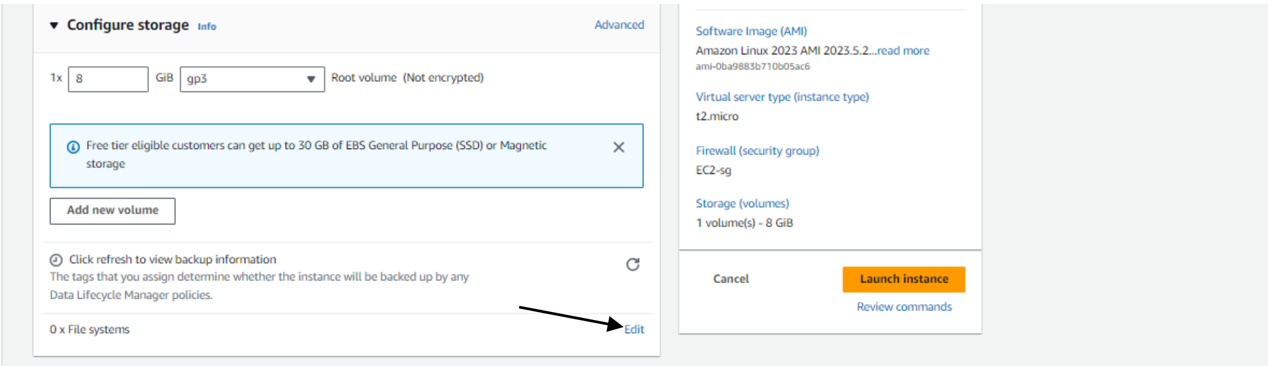


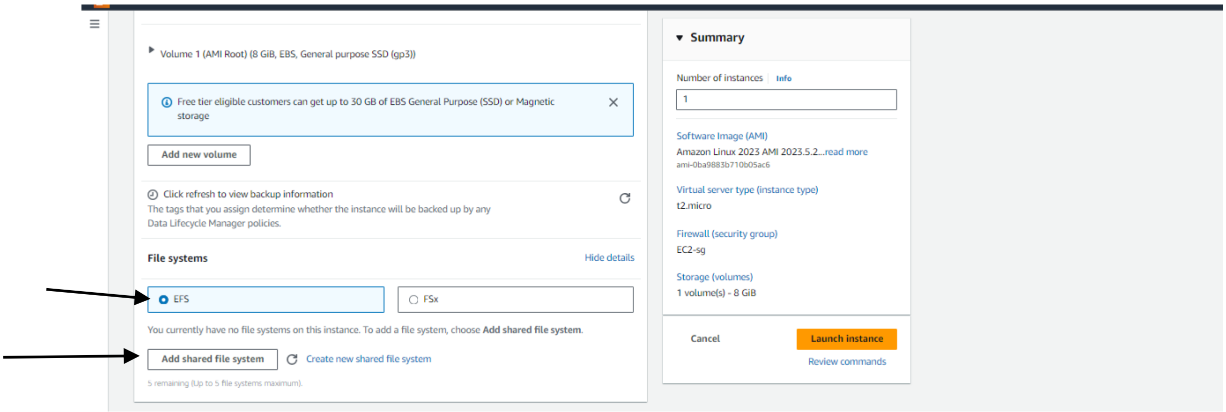


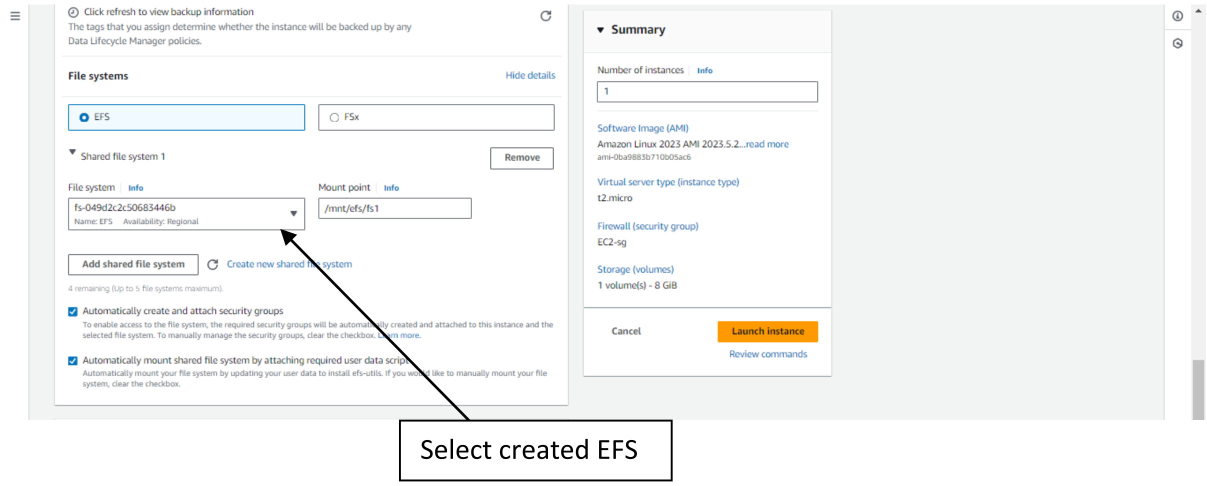
Select us-east-1a

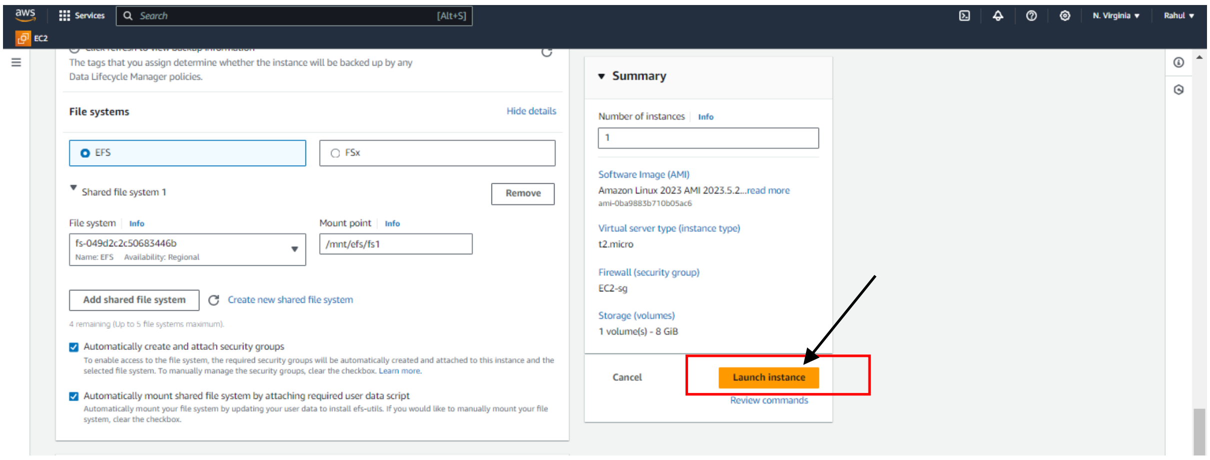







Volume 1 (AMI Root) (8 GiB, EBS, General purpose SSD (gp3))
Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
Hide details
EFS
FSx
You currently have no file systems on this instance. To add a file system, choose Add shared file system.
Add shared file system
Create new shared file system
Summary
Number of instances
Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
EC2-sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Review commands


Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
Hide details
EFS
FSx
Shared file system 1
Remove
File system
Info
fs-049d2c2c50683446b
Name: EFS Availability: Regional
Mount point
Info
/mnt/efs/fs1
Add shared file system
Create new shared file system
Automatically create and attach security groups
Automatically mount shared file system by attaching required user data script
Summary
Number of instances
Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
EC2-sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Review commands
Select created EFS


aws
Services
Search
[Alt+S]
N. Virginia
Rahul
EC2
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
Hide details
EFS
FSx
Shared file system 1
Remove
File system
Info
fs-049d2c2c50683446b
Name: EFS Availability: Regional
Mount point
Info
/mnt/efs/fs1
Add shared file system
Create new shared file system
Automatically create and attach security groups
Automatically mount shared file system by attaching required user data script
Summary
Number of instances
Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
EC2-sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Review commands

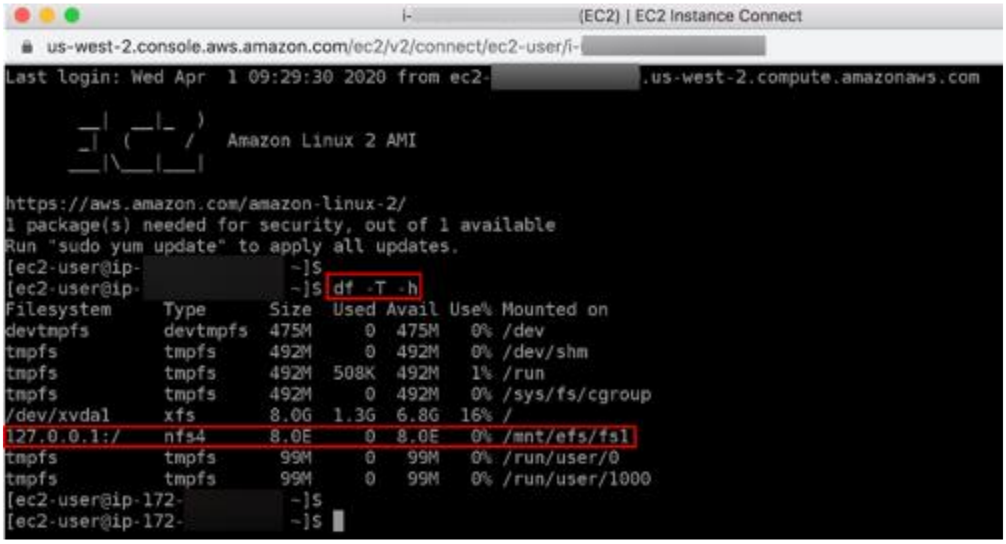
Verify EFS file system is mounted

Navigate to the EC2 console. In the EC2 navigation pane, choose **Instances**, then select the EC2 instance you created in Step 3. Choose **Connect**.

In the Connect to your instance dialog box, choose **EC2 Instance Connect** (browser-based SSH connection) and then choose **Connect**.

type the following command:

df -T –h



You can see that the EFS File System is mounted at mnt/efs/fs1 which is the same default path as mentioned in Step 3.4. This verifies that your file system is successfully mounted on the EC2 instance.

Step 5: Clean up

Terminate EC2 instance

5.1 — Open the Amazon EC2 console.

5.2 — In the navigation pane, choose Instances.

5.3 — Select the instance you created for this tutorial, and choose Actions, Instance State, Terminate.

5.4 — Choose Yes, Terminate when prompted for confirmation.

Note: This process can take several seconds to complete. Once your instance has been terminated, the Instance State *will change to* terminated *on your EC2 Console.*

Delete EFS file system

5.5 — Open the Amazon EFS console.

5.6 — On the File systems page, select the file system you created for this tutorial.

5.7 — Choose Delete.

5.8 — In the Delete file system dialog box, enter the file system ID shown, and choose Confirm to confirm the delete.

Also Delete Security groups.